

SkillCert - POC Functional Specification v1.2

Reconciled baseline - 4 October 2026

This edition reconciles the frozen v1.1 specification with the subsequent confirmed SkillCert decisions and its companion relational/domain model. The original v1.1 PDF is preserved. Where this document identifies a recommended technical default, it is not an additional confirmed business requirement.

Technology: .NET / EF Core / PostgreSQL; React / MUI. Source record: AFA Skills Record - Revised v3.

1. Purpose

SkillCert is a digital competency-record application intended to replace paper-based practical skills records such as the supplied AFA Skills Record. The source record organizes practical skills into hierarchical sections and records completion against individual assessable skills.

The POC provides configurable competencies and lists, individual competency currency and recertification, candidate-driven sign-off, physical signature capture, optional authenticated reviewer confirmation, user/group assignment, scheduled opportunities, instructor-led group attestation, administrative reporting, and automatic archival of completed training-record PDFs.

The POC deliberately excludes notifications, list revision tracking, and other advanced certification workflows.

2. Competencies

A Competency is one independently assessable skill, with a persistent identity independent of revisions and competency lists. It has Id, Code, CurrentRevisionId and IsActive. A competency may be reused across lists but may occur only once within each list. The server resolves the current revision for assessment; clients do not choose historical revisions.

3. Competency Revisions

A CompetencyRevision contains CompetencyId, RevisionNumber, Title, optional Description, optional RecertificationDays, publication metadata, resources, permitted review methods and InvalidatesPreviousReviews. Historical revisions are retained for audit and never used for new assessments.

Resources

A revision may contain zero or more URL resources with title, URL and type. Initial types are Web Page, Video and Document. Uploaded resource management is outside scope.

Recertification

RecertificationDays is a nullable positive integer. NULL means no time-based expiry. Expiry is ReviewedAt plus the stored number of elapsed 24-hour days. Human labels such as "yearly" for 365 days are display conveniences; they do not change arithmetic to calendar years. New non-breaking revisions do not change the interval attached to existing reviews.

Administrative edits

Administrators may correct minor editorial errors in assessed revisions without publishing a new revision. This does not invalidate reviews or require reassessment. Important configuration edits are recorded in a lightweight conventional audit log. Existing PDFs remain unchanged; a historical revision query shows corrected text, with the audit log providing the change record.

Publishing revisions

Substantive policy/procedure changes use "Publish new revision". The administrator explicitly chooses whether earlier reviews remain acceptable. Non-breaking publication preserves earlier currency. Breaking publication invalidates reviews against earlier revisions from PublishedAt. Recertification-policy changes use a new revision. Only the newest/current revision may receive new assessments; the server resolves it at sign-off. Historical assessments remain evidence even after invalidation.

4. Competency Lists

A CompetencyList references competencies from the shared library. Lists have recursively nested headings and explicit ordering; codes do not determine order. One unified CompetencyListNode tree represents both heading nodes and competency leaves. Parents must belong to the same list; cycles are prohibited. Heading nodes can contain headings or competencies; competency nodes cannot have children.

List membership

The database enforces uniqueness of (CompetencyListId, CompetencyId) for competency nodes. A competency may appear in many lists but only once in each list.

List revisions

List revision tracking is outside the POC. Changes are live. Compliance evaluates the current tree, and archived PDFs preserve the hierarchy rendered at generation. List changes never overwrite competency review history.

5. Users

Every system user can act as a candidate; there is no separate Candidate entity. Authentication is separate from the domain model. Users should generally be deactivated rather than deleted where historical reviews reference them.

6. User Groups

Users may belong to zero or more administrator-configured assignment groups, such as New Patroller, Returning Patroller and Senior Patroller. Groups are not mutually exclusive. Administrators assign competency lists to groups and users to groups.

A user's required lists are the distinct union inherited through group memberships. There are no direct user/list assignments in the POC; exception requirements use an administrator-configured group. Shared competencies remain one requirement regardless of how many lists or group paths introduce them.

7. Reviewer Classifications

Reviewer authority is separate from assignment groups and administrative permissions. Self and Peer are review methods; Instructor and Supervisor are seeded reviewer classifications held by system users. Classifications are DB-configurable, including affirmation policy. There is no POC UI for creating/editing classification definitions; administrators can assign/remove users from classifications.

Initial policies: Instructor uses Automatic; Supervisor uses Reviewer Confirmation. An authenticated Instructor assessment normally starts NotRequired. A candidate naming a Supervisor creates a Pending claim. Peer and Self require no signature or later confirmation and are immediately effective, starting NotRequired. Every competency defines the review methods it permits.

8. Competency Reviews

Each assessment creates one independent CompetencyReview for one candidate and competency revision. Fields: Id, CandidateUserId, CompetencyId, CompetencyRevisionId, Outcome, ReviewedAt, ReviewerUserId, optional ReviewerClassificationId, ReviewerName snapshot, optional ReviewSignatureId, optional Comment, optional OpportunityId, ConfirmationStatus, optional ConfirmedAt/ConfirmedByUserId, optional RejectedAt/RejectedByUserId/RejectionReason, CreatedAt and CreatedByUserId.

Outcomes are Competent and NotCompetent. ReviewerUserId is required for every review method. Assessment evidence is immutable; later assessments create new rows. Confirmation is a one-time lifecycle decision recorded on the row. ReviewEvent and CorrelationId are not part of the POC. Review voiding/correction workflows are outside scope.

9. Sign-Off Basket

The sign-off basket is UI-only, not a persistent domain entity. Candidates add competencies and present their device for sign-off. One interaction creates an independent CompetencyReview per selected competency in one transaction, optionally sharing a ReviewSignature and server-generated ReviewedAt. No event/batch identity is stored or reconstructed.

Review compatibility

Every competency in the basket must permit the selected method/classification. Duplicate candidate/competency pairs within a submission are rejected. The server resolves the newest revision at submission.

Reassessment

Currently valid competencies may be included for reassessment; currency does not prevent basket selection.

10. Reviewer Identity and Signature

All reviewers are registered system users. Store ReviewerUserId and a historical ReviewerName snapshot on each review.

Instructor and Supervisor

Select users holding the relevant classification. Candidate-device classified sign-off captures a physical touchscreen signature. The signature is associated evidence, not cryptographic identity verification. An automatic Instructor assessment requires the Instructor to authenticate; selecting a name alone does not establish authenticated automatic authority.

Peer and Self

Peer uses another registered user, not an external manually named reviewer. Self uses the candidate's own user ID. Neither requires a signature or later confirmation; both are immediately effective if permitted by the competency.

Shared signature storage

ReviewSignature contains Id, Data, ContentType and CreatedAt. Multiple reviews created in one interaction reference the same ReviewSignatureId, storing the image once. Group attestation may also share a signature across many reviews. No parent ReviewEvent or CorrelationId is introduced. Global hash-based deduplication across different interactions is not required.

11. Reviewer Confirmation

Confirmation states are NotRequired, Pending, Confirmed and Rejected. Only NotRequired and Confirmed reviews participate in currency.

Approval queue

Authenticated reviewers see pending claims naming them. The UI may visually group rows by candidate, reviewer, ReviewedAt and ReviewSignatureId. This grouping has no domain significance. The reviewer confirms or rejects the presented group; partial approval is outside the workflow. Group decisions commit atomically.

Only the named reviewer may confirm/reject. Administrator status does not permit impersonation. Removing a classification after claim creation does not remove the named reviewer's right to decide that existing claim, but prevents creation of new reviews under that classification. Reject requires a reason; rejected evidence is retained and never becomes effective. A rejected review cannot return to Pending.

Achievement date

Confirmation accepts evidence retrospectively at ReviewedAt; it never moves achievement or expiry to confirmation time. October 4 achievement confirmed October 18 remains achieved October 4, and any training-record completion established by it uses October 4.

12. Competency Currency

Currency is derived for User + Competency, independent of lists. Status values are NotCertified, Current, Expired and NotCompetent. Reasons distinguish NeverReviewed, Current, RecertificationExpired, RevisionInvalidated and NotCompetent. HasPendingReview is separate from status.

Select evidence

Use accepted NotRequired/Confirmed reviews with ReviewedAt at/before the evaluation time. Latest ReviewedAt wins, with CreatedAt and Id as deterministic tie-breakers. Do not fall back to an older favourable review. Pending/rejected reviews do not displace accepted evidence.

Competent and expiry

Competent establishes achievement at ReviewedAt. Use RecertificationDays from the assessed revision. No interval means no time-based expiry. The competency is Current strictly before ExpiresAt and Expired at/after that timestamp, unless a later assessment or breaking revision invalidates it.

Breaking revision

A breaking revision invalidates older-revision competency evidence from PublishedAt. Return NotCertified with RevisionInvalidated reason for invalidated competent evidence. Historical reviews remain intact. A late confirmation does not bypass a newer breaking revision.

Not Competent

An effective NotCompetent assessment removes existing currency from ReviewedAt. Only a later effective Competent assessment restores it. Pending NotCompetent has no effect until accepted.

Historical queries

Historical evaluation asks what currently accepted evidence establishes about the requested time, not what the application knew at that time. Knowledge-time/event-sourcing history is outside scope.

13. Candidate Experience

The candidate-facing application presents assigned competency lists while preserving their configured hierarchy. A competency displays current status, relevant dates, description/resources, and available review methods.

Useful presentation states include Not Certified, Current, Expired, Not Current - Not Competent, and Pending Confirmation. Expiring Soon may be shown as presentation information but is not a separate currency state.

- Inspect competencies and access resources.
- View personal review history and currency.
- Add competencies to the sign-off basket.
- Reassess currently valid competencies.
- Register for opportunities.
- View/download training records.

14. Opportunities

An Opportunity is a scheduled training/assessment event with Title, optional Description and free-text Location, StartsAt, EndsAt, optional Capacity, Status, AudienceMode and creation metadata. Administrators create/configure opportunities; instructor-created events are outside scope. Opportunities are retained, never hard-deleted in the POC.

Each opportunity has one or more competencies and assigned instructors, audience configuration and candidate registrations. Audience is AllUsers or members of selected administrator-configured groups; eligibility is checked at registration.

Opportunity competencies reference Competency, not a revision. Resolve the newest/current revision when assessing. Initial population from a list copies distinct competency IDs and creates no permanent list dependency. A competency occurs only once per opportunity.

Multiple instructors may be assigned. The performing reviewer is recorded on each assessment, with the Instructor classification held at assessment time. Historical assessments survive later removal of that classification. Assigned instructors handle operational delivery.

15. Opportunity Registration and Attendance

Candidates discover opportunities in the application and register through listings or opportunity QR. QR directs candidates into registration/check-in so they populate the instructor's participant list. QR uses a signed/opaque token rather than a persistent domain entity; it does not bypass audience or capacity rules.

OpportunityRegistration relates one registered user to one opportunity. States are Registered, Attended and Withdrawn, with only one registration per user/opportunity. Assigned instructors manage attendance; administrators may override attendance. Only Attended candidates are included in attestation by default.

Configured capacity is a hard limit. There is no waitlist. Recommended seat-count default: Registered and Attended occupy capacity; Withdrawn releases it. Check capacity transactionally. Registration windows, reactivation and attendance-correction details are implementation defaults to finalize. Payments, recurring scheduling and calendar integration are outside scope.

16. Group Attestation

Group attestation allows an authenticated assigned Instructor to assess classroom-scale candidate/competency pairs from their own device. Thirty candidates and thirty competencies can produce 900 independent reviews in one transaction, avoiding repeated signing on individual candidate devices.

The UI may initially select all attended candidates x all opportunity competencies and allow exceptions. Attested/Competent creates a Competent review. Not Assessed creates no review. Not Competent must be explicitly selected; deselection never implies incompetence.

Reviews may share exact server-generated ReviewedAt, reviewer identity/classification, OpportunityId and ReviewSignatureId. Each remains atomic. There is no CorrelationId, ReviewEvent or business logic reconstructing a batch. Opportunity context is provenance, not a continuing condition of currency. Automatic authenticated Instructor assessments are immediately effective.

17. Opportunity Lifecycle

Explicit transitions are Draft -> Open -> InProgress -> Completed, and Open -> Cancelled. Status is not inferred from the clock. Cancellation of an unused draft is a recommended edge-case default, not an additional confirmed transition.

Administrators control definition/configuration. Assigned instructors view participants, manage attendance, display/use check-in QR, attest and mark Completed. Completion means event delivery has ended; it does not imply every participant was assessed or competent. Completed opportunities are historical/read-only in normal instructor workflows. Opportunities remain retained historically.

18. Authorization

Capabilities are additive rather than mutually exclusive roles.

Authenticated User

- View own competency requirements, history, and currency.
- Access competency resources.
- Initiate sign-off.
- Register/check into opportunities and view own opportunities.
- Export own current training record.

Instructor Classification

- Access assigned opportunities.
- View participants and manage attendance.
- Perform group attestation.
- Complete assigned opportunities.

Instructor status does not automatically provide organization-wide access to candidate training records.

Supervisor Classification

- Access confirmation requests naming them.
- Inspect associated reviews/signature.
- Confirm or reject those requests.

Administrator

- Manage users, groups, group/list assignments, competencies/revisions, competency lists, user/classification assignments and opportunities.
- Access reporting and administrative audit information.
- View organization-wide training records.

Administrators cannot directly rewrite competency review evidence.

19. Administration

The POC administration area includes Dashboard/Reporting, Users, Groups, Competencies, Competency Lists, Reviewer Classification Assignments, Opportunities, and Audit.

Competency management explicitly separates Edit Current Revision from Publish New Revision. Publishing a new revision requires the administrator to choose whether it invalidates existing currency. List administration provides a hierarchical tree editor with explicit ordering and competency-library selection.

20. Reporting

Reporting is included in the POC, focused on currency and training gaps. Initial reports are:

Compliance Overview: organization/group-level compliance; a user is compliant with a list only when every required competency is Current.

User Compliance: candidates with missing, expired or otherwise non-current competencies.

Competency Compliance: currency rates by competency to identify organization-wide skill gaps.

Recertification Forecast: competencies approaching expiry over configurable time windows.

Pending Approvals: outstanding confirmation claims and responsible reviewers.

Individual Training Record: a candidate's current competency record and supporting review history.

Review activity reporting may be added if POC development time permits.

21. Training Record PDF

Training-record PDFs resemble the supplied paper AFA Skills Record, preserving hierarchy and compact date/evaluator styling. Include candidate identity, list title, completion date, complete hierarchy as rendered at generation, competency codes/titles, effective sign-off dates and evaluator information from reviewer-name snapshots.

The PDF excludes competency expiry dates, physical signature images, descriptions/resources and detailed historical reviews. Those remain available through application data where authorized.

Compliance

A nonempty list is compliant only when every distinct competency currently contained in it is Current. Empty lists are non-compliant. No optional/alternative competency rules apply. Administrators may evaluate an unassigned list, but automatic required-list processing uses group-inherited assignments.

Completion date

Use competency evidence, not confirmation, assignment or rendering time. Preserve the established completion date while nightly-observed compliance continues. For newly observed completion, use the beginning of the evidence-supported simultaneous-current period under the current list. For the simplified requirement-removal case, remaining competencies achieved September 1 and a missing requirement removed October 4 yield September 1 completion. Historical list-change reconstruction is outside scope.

22. Automatic Training Record Archival

The compliance engine runs nightly. First observed assigned compliance and an observed NonCompliant-to-Compliant change trigger automatic PDF archival. Remaining compliant does not repeatedly generate PDFs. Intermediate changes that reverse between nightly runs are intentionally ignored; adding and removing a requirement during the day does not force recompliance.

TrainingRecord has Id, UserId, CompetencyListId, CompletionDate, GeneratedAt, BlobPath, Sha256Hash and Trigger. Trigger is ComplianceAchieved or AdministratorManual. CompletionDate is evidence-based; GeneratedAt is actual archival time. Candidate downloads of the current record do not create archive snapshots or suppress automatic archival.

An operational checkpoint may track prior nightly observations without becoming authority for current compliance. Automatic generation must be idempotent; retries must not create duplicate archives. The domain model recommends uniqueness by user/list/completion date for automatic records. Manual archives are separate.

Historical list structure

The immutable PDF preserves the list and displayed evidence as rendered. Future list edits, editorial corrections or user-name changes never alter it. The document must be understandable independently of the database. Erroneous/voided/superseded archive handling is outside scope. Failed generation/upload is retryable and must not be represented as a completed archive.

23. Audit Principles

Competency review assessment evidence is historical and immutable; later reviews do not overwrite earlier assessments. One-time confirmation/rejection is explicitly permitted, retaining actor, time and rejection reason.

Important administrative configuration changes are auditable, including editorial revision corrections, publication, group/membership/list assignment and opportunity configuration. A lightweight conventional audit log stores actor, timestamp, action, affected entity and relevant before/after values. Event sourcing, ReviewEvent and CorrelationId are unnecessary. Audit visibility is administrative. Review voiding/correction and archived-record correction are outside scope; voiding is noted as a future enhancement.

24. Explicitly Outside POC Scope

- Notifications/email reminders.
- Competency-list revision tracking.
- Historical list-completion auditing beyond archived PDFs.
- List-level certification expiry.
- Complex optional/alternative competency requirements.
- Certificates separate from training records.
- Instructor-created opportunities.
- Waitlists, payments, recurring opportunity scheduling, and calendar integrations.
- Complex competency prerequisites.
- Uploaded training-resource management.
- Cryptographic signature verification or advanced electronic reviewer identity verification.
- Archived-record correction/voiding.
- Automated recommendation/matching of opportunities.
- Multi-organization tenancy unless separately added to scope.

25. POC Domain at a Glance

User -> group memberships -> group/list assignments -> competency-list node tree -> competencies.

User -> reviewer classifications, candidate/reviewer competency reviews, opportunity registrations and training records.

Competency -> retained revisions -> URL resources, day-based recertification and permitted methods/classifications.

Opportunity -> competencies, assigned instructors, audience groups and registrations/attendance.

CompetencyReview -> candidate, newest revision at assessment, registered reviewer, reviewer-name snapshot, outcome, ReviewedAt, optional shared signature/opportunity and confirmation decision.

Technology direction remains .NET / EF Core / PostgreSQL and React / MUI, with blob storage for archived PDFs. Authentication is separate from domain identity. The companion relational/domain document defines keys, relationships, constraints, indexes and derived rules. Technical defaults remain labelled; this reconciliation does not add new product scope.

26. CSV Competency Import

CSV import is an Administrator-only, create-only function for loading new competencies and initial revisions. It never updates existing competencies or publishes additional revisions of existing competencies. List hierarchy/membership import is outside this feature.

Fields

Support competency code, title, optional description, nullable positive RecertificationDays and permitted methods. Resource title/URL may be imported when represented by the agreed format. Exact resource/method encoding remains a file-contract detail. A representative header is:

Code,Title,Description,RecertificationDays,SelfReview,PeerReview,InstructorReview,SupervisorReview

An initial row can use 365 for yearly rather than the old calendar phrase "1 year".

Validation and preview

Validate the entire file before import; show rows to create, warnings and row-specific errors. Invalid rows block final import until corrected; there is no silent partial import. Detect normalized duplicate codes. Existing codes, including inactive competencies, are conflicts. Recommended simple default: conflicts block the file, rather than silently skipping rows; an explicit skip-with-preview option is not required by this POC. Concurrency is guarded by the database code uniqueness constraint.

Application

Create accepted competencies and initial revisions atomically. Finalize initial publication during implementation. Updates, editorial edits and additional revisions of existing competencies are outside CSV scope.